IP-GCA: An Implicit Preference-Gated AI Copilot Architecture

for Real-Time Human-in-the-Loop Customer Support via Dynamic Execution Graphs and MCP

Hans Alfonso Mendoza Alva · UTEC

1. Problem Statement

Enterprise contact centers face a critical architectural trade-off: rigid decision-tree systems guarantee rule compliance but fail on complex user intents, whereas fully autonomous Large Language Models (LLMs) offer fluency but introduce severe operational risks, including factual hallucinations, unconstrained database mutations, and context degradation. While generative AI assistance improves agent resolution productivity by 14–15% [1], existing systems treat Retrieval-Augmented Generation (RAG) [2, 6, 8], agentic tool calling [10, 9, 11], and human feedback as disconnected, static pipelines. Current architectures lack a unified controller that dynamically routes queries based on context uncertainty and latency budgets, nor do they natively integrate human operator edit deltas into the generation mechanism in real time under latency bounds.

2. Research Question & Hypothesis

Research Question: How can a dual-loop Implicit Preference-Gated Copilot Architecture (IP-GCA), unifying uncertainty-aware dynamic routing, edit-delta preference conditioning, and Model Context Protocol (MCP) tool isolation minimize factual hallucinations ($< 2\%$) and maximize generation faithfulness (> 0.90 RAGAS), while maintaining real-time interaction bounds in a Human-in-the-Loop (HITL) customer support environment?

Hypothesis: An integrated IP-GCA architecture featuring a Gated Preference-Latency Controller (GPL-Controller) that dynamically selects execution paths and conditions generation prompts via low-rank operator preference keys ($\mathbf{k}_{\text{pref}}$) derived from UI edit deltas achieves significantly higher faithfulness (> 0.90) and tool parameter precision ($> 92\%$) than static Naive RAG and unconstrained agent pipelines, while enforcing latency bounds.

3. Key Architectural Contributions

This thesis introduces three core Computer Science contributions embedded directly within the proposed system architecture:

1. **Dual-Loop Preference-Gated Architecture (IP-GCA):** A system architecture featuring a *Gated Preference-Latency Controller (GPL-Controller)*. Unlike static pipelines, the GPL-Controller dynamically evaluates a joint context-uncertainty and latency-budget vector $(\mathbf{u}_{\text{ctx}}, \mathbf{l}_{\text{budget}})$ to route queries along a variable-depth execution graph (Fast-path parametric to Sparse-Dense Hybrid RAG to Agentic Reflection to Abstention).
2. **Edit-Delta Preference Conditioning Mechanism:** An in-context architectural bridge that captures semantic edit deltas (Δ_{edit}) between raw AI suggestions (S_{gen}) and final human operator modifications (S_{human}). It automatically converts operator corrections into dynamic attention preference keys ($\mathbf{k}_{\text{pref}}$) injected into the GPL-Controller's working memory, enabling continuous online adaptation without full model retraining.
3. **Deterministic State Machine MCP Guardrail Layer:** An architectural isolation layer placing a Finite State Machine (M_{fsm}) between the non-deterministic LLM tool reasoning engine and external MCP JSON-RPC servers, ensuring all tool payloads strictly conform to parameter schema boundaries before triggering human UI verification.

4. Scope and Explicit Limitations

- **Models:** Instruction-tuned open-weights models specialized in tool calling and reasoning (Qwen-3-14B-Instruct, Gemma-4-31B, or DeepSeek-R1-Distill-Qwen-32B via 4-bit AWQ/vLLM execution); DeBERTa-v3-large (NLI entailment) and Prometheus-2 (8B) serve as local reference-free evaluators.
- **Protocol:** Tool execution is strictly isolated via MCP (JSON-RPC 2.0). Custom REST/GraphQL wrappers are excluded.
- **Compute:** Single workstation GPU (RTX 4090, 24 GB VRAM) leveraging vLLM accelerated inference and quantized model loading.
- **Modality:** Text-based customer support channels (live chat, tickets). Speech/telephony pipelines are out of scope.

5. Objectives

General Objective: Design, formalize, implement, and quantitatively evaluate the IP-GCA architecture to optimize generation faithfulness, dynamic tool execution, and online human preference adaptation under real-time latency constraints in customer support.

Specific Objectives:

1. Formalize the mathematical state transitions and routing algorithms of the *GPL-Controller* under joint latency and uncertainty constraints.
2. Construct a two-stage hybrid retrieval engine (SPLADE + BGE-M3 + Cross-Encoder) and benchmark its Recall@K and MRR@10 against dense-only baselines.
3. Develop the edit-delta conditioning bridge ($\Delta_{\text{edit}} \rightarrow \mathbf{k}_{\text{pref}}$) to dynamically update prompt key-value caches from human operator edits.
4. Quantitatively evaluate the complete IP-GCA system across 4 dataset tiers using an 8-variant ablation matrix (A–H).

6. Formal Metrics & Non-Circularity

Let S_{gen} be the AI suggestion, S_{human} the operator-edited text, C the retrieved context, and P the tool payload:

- **Faithfulness (F_{faith}):** Proportion of atomic claims in S_{gen} directly entailed by context C:

$$F_{\text{faith}}(S_{\text{gen}}, C) = \frac{|\text{Claims}(S_{\text{gen}}) \cap \text{SupportedBy}(C)|}{|\text{Claims}(S_{\text{gen}})|}$$

- **Operator Alignment Index (A_{op}):** Normalized semantic similarity and Levenshtein edit distance reduction between S_{gen} and S_{human} over interaction sequence t:

$$A_{\text{op}}(t) = 1 - \frac{\text{EditDist}(S_{\text{gen}}^{(t)}, S_{\text{human}}^{(t)})}{\max(|S_{\text{gen}}^{(t)}|, |S_{\text{human}}^{(t)}|)}$$

- **Tool Invocation Precision (A_{tool}):** AST-based parameter match score between generated payload P_{gen} and ground-truth execution payload P_{gt}:

$$A_{\text{tool}} = \mathbb{I}(\text{Tool}_{\text{gen}} == \text{Tool}_{\text{gt}}) \times \frac{|\text{Params}(P_{\text{gen}}) \cap \text{Params}(P_{\text{gt}})|}{|\text{Params}(P_{\text{gt}})|}$$

- **End-to-End Latency (L_{e2e}):** Total pipeline processing time: $L_{\text{e2e}} = t_{\text{routing}} + t_{\text{retrieval}} + t_{\text{rerank}} + t_{\text{generation}} < 1200$ ms.

Non-Circularity: All metrics are computed using an independent judge model (DeBERTa-v3 / Prometheus-2) completely decoupled from generation models.

7. Expected Results & Ablation Matrix

Goal	Metric	Expected Outcome
Factual Grounding	F_{faith}	> 0.90 score; $< 2\%$ hallucination rate.
Retrieval Precision	MRR@10, P_{ctx}	MRR@10 > 0.82; $+12\%$ gain over dense-only.
Operator Alignment	A_{op}	$> 30\%$ reduction in human edit distance over 500 session steps.
Tool Execution	A_{tool}	$> 92\%$ tool parameter precision via MCP + FSM.
Real-Time Latency	L_{e2e}	$<$ 1200 ms for routed single-step; $<$ 2200 ms multi-hop.

Ablation Variants (A–H): **A**: Naive RAG (Dense vector search). **B**: A + SPLADE sparse expansion. **C**: A + Cross-Encoder reranking. **D**: Hybrid RAG (Dense + Sparse + Reranker). **E (Full IP-GCA)**: D + GPL-Controller + Edit-Delta Bridge + MCP Guardrail + HITL. **F**: E without Edit-Delta Preference Keys ($\mathbf{k}_{\text{pref}}$). **G**: E without dynamic GPL-Controller routing (fixed pipeline). **H**: E with randomized context (control).

8. Datasets

- **Primary (RAG & Reasoning):** *HotpotQA* / *MS MARCO* for multi-hop retrieval and hallucination evaluation.
- **Secondary (Agentic Tools):** *Berkeley Function Calling Leaderboard (BFCL)* / *AgentBench* for MCP tool precision.
- **Domain (Customer Support):** *Bitext Customer Support Dataset* for intent extraction, simulated operator edits, and dialogue latency.
- **Synthetic (Enterprise PDF):** *RAGAS Testset Generator* applied to real-world corporate manuals for citation grounding.

References

[1] E. Brynjolfsson, D. Li, L.R. Raymond, "Generative AI at Work," *QJE*, 140(2), 889–942, 2025.

[2] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *NeurIPS*, 2020.

[3] T. Formal et al., "SPLADE: Sparse Lexical and Expansion Model for Information Retrieval," *ACM SIGIR*, 2021.

[4] R. Nogueira, K. Cho, "Passage Re-ranking with BERT," *arXiv:1901.04085*, 2019.

[5] O. Khattab, M. Zaharia, "ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction," *ACM SIGIR*, 2020.

[6] A. Asai et al., "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection," *ICLR*, 2024.

[7] S.-Q. Yan et al., "Corrective Retrieval Augmented Generation (CRAG)," *ICML*, 2024.

[8] S. Jeong et al., "Adaptive-RAG: Learning to Adapt Retrieval-Augmented LLMs," *NAACL*, 2024.

[9] N. Shinn et al., "Reflexion: Language Agents with Verbal Reinforcement Learning," *NeurIPS*, 2023.

[10] T. Schick et al., "Toolformer: Language Models Can Teach Themselves to Use Tools," *NeurIPS*, 2023.

[11] C. Packer et al., "MemGPT: Towards LLMs as Operating Systems," *arXiv:2310.08560*, 2023.

[12] Anthropic PBC, "Model Context Protocol (MCP) Specification," *modelcontextprotocol.io*, 2024.

[13] Shahul ES et al., "RAGAs: Automated Evaluation of Retrieval Augmented Generation," *EACL*, 2024.

[14] X. Liu et al., "AgentBench: Evaluating LLMs as Agents," *ICLR*, 2024.